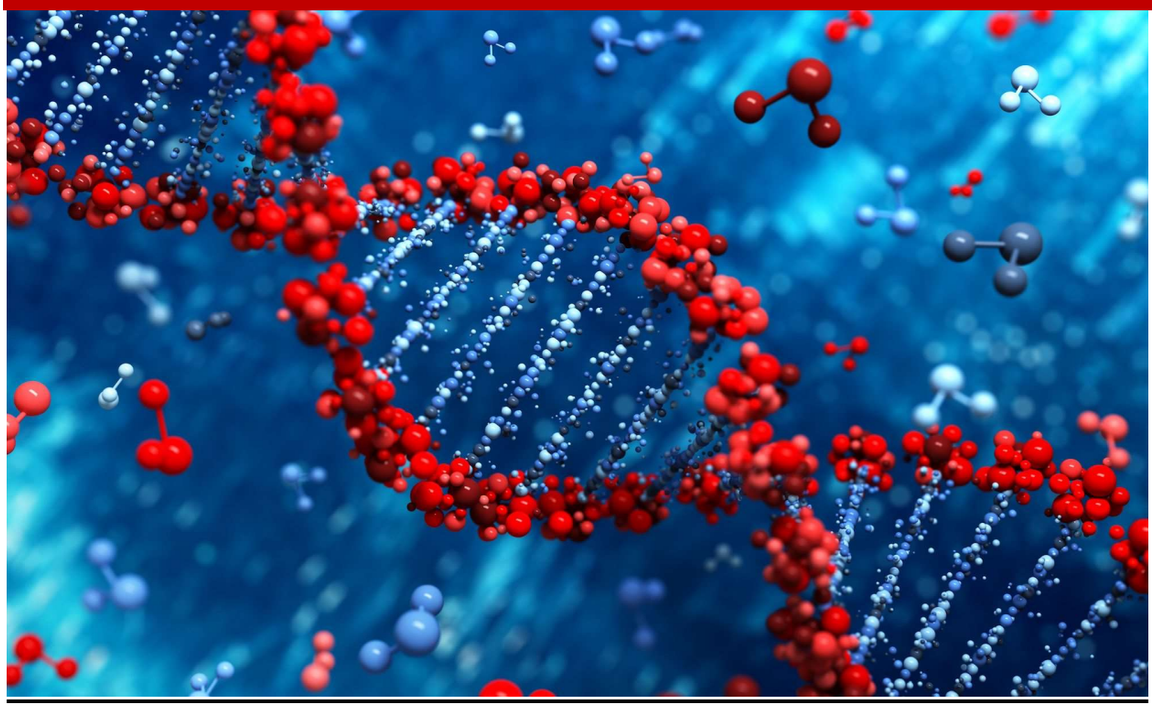
BIO203 - Methods in Molecular Biology

Mid-Term File 2023 By

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Microbiology application:

The applications of microbiology including:

1= Identification of bacteria: 2= Slow growing bacteria 3= Bacterial antibiotic resistance genes. 4= VIRUS: almost all viruses identification and viral load. 5= FUNGUS 6= Aspergillus spp 7= Candida spp 8= Plasmodium spp 9= Trypanosoma spp

Forensic Application of PCR:

Forensic applications: 1= DNA finger printing 2= Detection of Mutations: Inherited disorders & carriers Track DNA abnormalities 3= Prenatal diagnosis of genetic disorders. 4= Detection of pathogens Pre-natal diagnosis DNA fingerprinting Gene therapy Mutation screening 5= Drug discovery 6= Classification of organisms Genotyping.

Nested PCR differ Multiplex PCR:

Nested PCR: in this type of PCR the Two pairs (instead of one pair) of PCR primers are used to amplify a fragment. First pair amplify a fragment similar to a standard PCR, and have Very low probability of nonspecific amplification.

Multiplex PCR: Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using more than one pair of primers. The use of multiple, unique primer sets within a single PCR reaction to produce amplicons of varying sizes specific to different DNA sequences.

Type 1 of Restriction and Modification (R-M):

It is characterized from E. coli K 12. According to this The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit. The Type I systems are of little value for gene manipulation.

Uses of Asymmetric PCR:

It is used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal. It is also used to generate long primers and single-stranded DNA for incorporating cross-linking analogs into specific sites in a DNA probe.

Amplified fragment length polymorphism (AFLP) PCR:

Amplified fragment length polymorphism (AFLP) is a PCR-based technique that uses selective amplification of a subset of digested DNA fragments to generate and compare unique fingerprints for genomes of interest. The power of this method relies mainly in that it does not require prior information regarding the targeted genome, as well as in its high reproducibility and sensitivity for detecting polymorphism at the level of DNA sequence. Widely used for plant and microbial studies, AFLP is employed for a variety of applications, such as: to assess genetic diversity within species or among closely related species, to infer population-level phylogenies and biogeographic patterns, to generate genetic maps and to determine relatedness among cultivars. Variations of standard AFLP methodology have been also developed for targeting additional levels of diversity, like transcriptomic variation and DNA methylation polymorphism.

Negative control and blank reaction:

Negative control: The Controls for specificity of the amplification reaction contains all reagents and a DNA template lacking the target sequence is called negative control.

BLANK REACTION: The Controls for contamination contains all reagents except DNA template is called blanked Reaction.

Hot start PCR:

- This is a technique that reduces nonspecific amplification during the initial set up stages of the PCR.
- The technique may be performed manually by heating the reaction components upto the melting temperature (e.g, 95°C).
- Before adding the polymerase, specialized enzyme systems have been developed that inhibit the polymerase's activity at ambient temperature.
- This is done either by the binding of an antibody or by the presence of covalently bound inhibitors that only dissociate after a high-temperature activation step.
- DNA Polymerase- Eubacterial type I DNA polymerase, Pfu.

Differentiate between the each Step of PCR:

1= Denaturation:

This is the First regular cycling event and consists of heating the reaction to 93°C - 98°C for 20-45 seconds. It causes melting of DNA template yielding single strands of DNA.

2= Annealing:

In this The reaction temperature is lowered to 40-65°C for 20-40 seconds.Annealing of the primers to the single stranded DNA template.

3= Extension:

At this step the DNA polymerase synthesizes a new DNA strand complementary to the DNA template strand by adding dNTP's. The temperature at this step depends on the type of DNA polymerase used. Taq polymerase has its optimum activity temperature at 70- 80°C. Commonly 72°C is used.

Methylation PCR (MSP):

Methylation-specific PCR is used to identify patterns of DNA methylation at CpG islands in genomic DNA. Methylation Specific PCR Methylation Specific PCR METHYLATION SPECIFIC PCR.

MSP is alos used in quantitative PCR provides information about methylation state of a given CpG island.

The Target DNA is first treated with sodium bisulfite, which converts unmethylated cytosine bases to uracil, which is complementary to adenosine in PCR primers.

Two amplifications are then carried out on the bisulfite-treated DNA: One primer set anneals to DNA with cytosines (corresponding to methylated cytosine), and the other set anneals to DNA with uracil (corresponding to unmethylated cytosine).

What are the Colony PCR, In Situ PCR, Asymmetric PCR:

Colony PCR: The screening of bacterial or yeast clones for correct ligation or plasmid products, and this type of PCR is called Colony PCR.

In Situ PCR: In Situ PCR (ISH) is a polymerase chain reaction that actually takes place inside the cell on a slide. In situ PCR amplification can be performed on fixed tissue or cells.

Asymmetric PCR: It is used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal. It is also used to generate long primers and single-stranded DNA for incorporating cross-linking analogs into specific sites in a DNA probe.

Nomenclature of The Smith and Nathan's (1973)

- A suitable system was proposed by Smith and Nathans (1973)
- The species name of host organisms is identified by the first letter of genus and first two letters of specific epithet.
- E. coli = Eco
- H. influenzae = Hin
- Strain identification is written as Eco K
- In case, host strain has several restriction and modification systems, these are identified by roman numerals, for example, in case of H. Influenzae.

Write the detiled note on Mini Primer PCR:

- Mini Primer PCR uses a thermostable polymerase (S-Tbr) that can extend from short primers as short as 9 or 10 nucleotides.
- This method permits PCR targeting to smaller primer binding regions, and is used to amplify conserved DNA sequences, such as the 16S (or eukaryotic 18S) rRNA gene.
- PCR that can extend from short primers

Conventional PCR (Real Time PCR or q-PCR):

In conventional PCR, the amplified DNA product, or amplicon, is detected in an end-point analysis. In real-time PCR, the accumulation of amplification product is measured as the reaction progresses, in real time, with product quantification after each cycle.

Ingredients:

- QRT-PCR methods use fluorescent dyes, such as Sybr Green, or fluorophore-containing DNA probes, such as TaqMan, to measure the amount of amplified product in real time.
- Q-PCR is commonly used to determine whether a DNA sequence is present in a sample and the number of its copies in the sample.

Difference between dNTPs and Assembly PCR:

dNTPs: Deoxynucleoside triphosphates (dNTPs) are the building blocks from which the DNA polymerase synthesizes a new DNA strand during successive cycles of PCR amplification. dNTPs consist of four basic nucleotides - dATP, dCTP, dGTP, and dTTP.The concentration of each dNTP (dATP, dCTP, dGTP, dTTP) in the PCR is usually 200 µM.

Assembly PCR: Assembly PCR used to assemble two or more pieces of DNA into one piece. Assembly PCR PRINCIPLE. Assembly PCR is the synthesis of long DNA structures by

performing PCR on a pool of long oligonucleotides with short overlapping segments, to assemble two or more pieces of DNA into one piece.

What reagents are used in standard PCR?

Generally, a complete polymerase chain reaction (PCR) requires five basic reagents, including 1= DNA template, 2= Forward and reverse primers, 3= DNA polymerase, 4= Deoxy nucleotide triphosphates (dNTPs) 5= Reaction buffer.

Symporters:

Symporters: Symporters are proteins that simultaneously transport two molecules across a membrane in the same direction.

Advantages of PCR:

- Specific
- Simple, rapid, relatively inexpensive
- Amplifies from low quantities
- Works on damaged DNA
- Sensitive
- Flexible

What are the Limitations of PCR:

1= Contamination risk 2= Primer complexities 3= Primer-binding site complexities 4= Amplifies rare species 5= Detection method.

DNA Polymerases sources:

- T. aquaticus
- Thermococcus litoralis
- Pyrococcus furiosus
- T. thermophilus

Define Alu-PCR and its uses:

The use of primers from a commonly repeated segment is called Alu-PCR.

Uses: it can help to amplify sequences adjacent (or between) these repeats.

Ligated pcr methodology:

Uses small DNA oligonucleotide 'linkers' (or adaptors) that are first ligated to fragments of the target DNA. PCR primers that anneal to the linker sequences are then used to amplify the target fragments.

Types of Restriction and modification (R-M) system:

Types of restriction and modification (R-M) system, At least four R-M systems are known

- Type I
- Type II
- Type III
- Type Iis

Define types of (R-M)

Type I: Type I systems were the first to be characterized from E. coli K 12. The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit.

Type II: Most of the useful R- M system is Type II. Type II enzymes recognize a defined sequence and cut within it

Type III: Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value.

Type Iis: Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness.

3 Steps of designing Primers:

There are 3 strategies for primer design: 1) insert-specific primers, 2) backbone-specific primers, and 3) orientation-specific primers

Number and size of restriction fragments:

The number and size of fragments generated by restriction enzyme depend on the frequency of occurrence of the target site in the DNA to be cut.

- Four base recognition site occurs every 44 (256) bp
- Six base recognition siteoccurs every 46 (4096) bp
- Eight base recognition site occurs 48 (65,536) bp